

Research Methodology

SLR to Technical Paper

Path to Technical Q1 Paper

By:

Dr. Zohair Ahmed



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Analysis of SLR Results

- **Summarize Key Findings:**
 - Group studies by method (e.g., RNNs, CNNs, Transformers).
 - Highlight leading datasets, metrics, and frameworks.
- **Identify Trends:**
 - Shift from statistical ML → deep learning → LLMs.
 - Increased use of multimodal datasets.
- **Detect Conflicts:**
 - Example: Contradictory findings in explainable AI results.
- **Spot Limitations:**
 - Some report SHAP effective; others find instability.
 - Small datasets, poor generalization, lack of real-world testing.
- **Output:** Analytical tables and graphs showing evolution or gaps.
- **Visual Suggestion:** Trend chart showing publication growth per year in a specific AI subdomain.

Identifying Research Gaps

- Finding gaps turns descriptive results into research opportunities.
- **Content Analysis:**
 - Examine “future work” or “limitations” sections in reviewed papers.
 - Note recurring unsolved issues.
- **Comparative Matrix:**
 - Cross-tabulate methods × datasets × evaluation metrics.
 - Identify blank cells = unexplored areas.
- **Bibliometric Tools:**
 - Use VOSviewer or CiteSpace for keyword co-occurrence.
- Discover emerging clusters or weakly connected topics.
- **Gap Typology:**
 - **Conceptual gaps:** no theory/model integration.
 - **Methodological gaps:** limited experimentation.
 - **Empirical gaps:** lack of large-scale evaluation.
 - **Contextual gaps:** underexplored domains (e.g., Urdu NLP).
- **Example:**

“In sentiment analysis, no study explores combining LLM prompt engineering with low-resource languages.”
- Gap identification table (methods vs datasets).



From Gaps to Research Questions

- Convert a gap into a testable and clear research question (RQ).
- **Transformation Process:**
- Identify problem area: *Few studies on multimodal sentiment fusion.*
- Convert to question:
 - RQ1: *How can attention-based cross-modal fusion improve accuracy in multimodal sentiment classification?*
- Ensure SMART characteristics:
 - **Specific** – clearly focused problem.
 - **Measurable** – linked to performance metrics.
 - **Achievable** – feasible with available data.
 - **Relevant** – adds value to the field.
 - **Time-bound** – can be achieved within your research timeline.
- **Checklist for RQ Quality:**
- Avoid yes/no questions.
- Link each RQ to the identified literature gap.
- Ensure theoretical and practical significance.

Defining Research Objectives & Hypotheses

- Translate RQs into clear, actionable goals.
- **Objectives:**
- Broad aim → “To design a multimodal LLM framework for cross-domain sentiment analysis.”
- Specific objectives:
 - Develop a vision-language embedding model.
 - Evaluate model on benchmark datasets (e.g., MOSEI, CMU-MOSEAS).
 - Compare performance with SOTA baselines.
- **Hypothesis Example (Quantitative):**
- H1: “Integrating image-text co-attention increases classification accuracy by $\geq 5\%$ over MAG-BERT.”
- Each hypothesis should be *testable and falsifiable*.



Literature Gap Validation

- Verify that your identified gap truly exists and is worth solving.
- **Methods:**
- **Pilot Study:** Run small-scale experiment with limited data.
- **Expert Validation:** Discuss with senior researchers or domain experts.
- **Cross-check with Databases:** Use *Papers with Code* and *Scopus* for latest works.
- **Trend Validation:** Ensure topic has upward trend (new citations, recent conference focus).
- **Example:**
Before proposing a new alignment model, confirm that no paper since 2023 has addressed it on arXiv or ACL Anthology.



Developing a Conceptual Framework

- A conceptual framework visualizes how variables and components interact.
- **Steps:**
 - Identify theoretical basis from SLR.
 - e.g., Cognitive Theory of Multimodal Learning.
 - Map relationships:
 - Input (text, image) → Process (fusion) → Output (sentiment).
 - Add moderating factors (e.g., dataset domain, noise).
- Justify using models from prior studies.
- **Example:**

A model integrating CLIP embeddings + transformer decoder for multimodal understanding.

Designing the Research Methodology

- Which methodology best answers your RQs?
- **Options:**
- **Experimental:** Build and test models, evaluate metrics.
- **Computational/Theoretical:** Develop or modify algorithms.
- **Empirical:** Analyze existing data or systems.
- **For AI/ML Projects:**
- **Data Sources:** Kaggle, Hugging Face, academic datasets.
- **Data Processing:** Cleaning, augmentation, annotation.
- **Model Design:** Choice of architecture (BERT, GNN, CNN).
- **Evaluation Metrics:** Accuracy, F1, MAE, BLEU, ROUGE.
- **Validation Techniques:** Cross-validation, Ablation studies.
- **Reproducibility Tools:** GitHub, OSF, or Jupyter Notebooks for open science.
- **Visual:** Flowchart of research methodology stages.



Drafting a Complete Technical Paper

- **Recommended Structure (Q1 Standard):**
- **Abstract:** concise purpose, method, results, and significance.
- **Introduction:**
 - Establish context from SLR.
 - State problem, motivation, and novelty.
- **Related Work:**
 - Summarize SLR insights relevant to your approach.
 - Show where your work fits.
- **Methodology:**
 - Detailed architecture, dataset, and experimental setup.
- **Results:**
 - Comparison with SOTA.
 - Tables, charts, ablation analysis.
- **Discussion:**
 - Interpret results, limitations, implications.
- **Conclusion and Future Work.**
- **Tip:**

Integrate SLR references throughout especially in the *Related Work* and *Discussion* sections.

Evaluating Novelty and Contribution

- **Checklist for Novelty:**

- Does it propose a new algorithm or modification?
- Is there a new combination of existing techniques?
- Are you solving a real-world or underexplored problem?
- Does your approach achieve measurable improvement?

- **Contribution Types:**

- **Theoretical:** new model or formulation.
- **Empirical:** improved performance or dataset.
- **Methodological:** novel workflow or framework.

- **Example:**

A new loss function for multimodal fusion that improves F1 by 3–5%.



Selecting Q1 Publication Targets

- **Journal Selection Strategy:**
 - Subject = AI, ML, Data Science.
- **Match *scope and method*:**
 - Method-heavy → *Information Fusion, Pattern Recognition*.
 - Application-heavy → *ESWA, KBS, Applied Soft Computing*.
 - NLP-focused → *Information Processing & Management, IEEE Access*.
- **Check Impact Indicators:**
 - CiteScore, SNIP, and Review Time.
- **Tip:**

Read 2–3 recently published articles in your target journal before submission.
- **Use Scopus/SJR filters:**
 - Quartile = Q1

Transition from SLR to Thesis or Proposal

- **For thesis proposal:**
 - Use SLR's gap analysis as rationale.
 - Convert objectives into research aims.
- **For technical paper:**
 - Focus on methods, experiments, and SOTA comparison.
- **Outcome:**

Stronger proposal defense, logical continuity, and publication potential.
- **Example:**

SLR identifies limited multimodal data fusion
- Thesis develops a novel transformer-based multimodal model.

Tools, Platforms, and AI Assistance

- **For Idea Generation:**
 - Research Rabbit, Connected Papers, ArXiv.
- **For Benchmark Validation:**
 - Papers with Code (SOTA tracking).
 - Semantic Scholar “influential citations.”
- **For Writing and Collaboration:**
 - Overleaf (LaTeX collaboration).
 - Zotero or Mendeley for references.
 - OSF or GitHub for open access data.
- **AI Writing Assistance:**
 - ChatGPT / Grammarly / Scite.ai for drafts, summaries, and citation checks.



Timeline / Workflow Overview

- **SLR → Identify Gaps**
 - Look at what's missing in literature (methods, datasets, evaluation, multimodality, etc.).
- **Gap → Research Questions & Objectives**
 - Formulate what exactly you want to solve.
- **Conceptual Model / Framework**
 - Sketch how your method will work conceptually.
 - Define inputs, outputs, key components, and their relationships.
- **Pilot Study** (*optional but highly recommended before full implementation*)
 - Small-scale testing to validate feasibility.
- **Full Implementation & Experiments**
 - Build complete model, run on full datasets, evaluate SOTA.



Why a Pilot Study is Needed

- A **pilot study** is a **mini-experiment** done **before full-scale implementation** to check:

Purpose	Explanation	AI/ML Example
Feasibility	Can your idea actually work on real data?	Test cross-modal alignment on 500 sample videos instead of 10,000.
Resource Estimation	Time, GPU, memory, storage needed	How long training takes per epoch; required GPU VRAM.
Hyperparameter Sensitivity	See if model is stable	Try learning rate variations on small dataset.
Data & Preprocessing Issues	Check data quality	Ensure dataset labels are consistent; detect missing features.
Initial Performance Check	Confirm approach beats baseline	Compare small-scale performance vs baseline model.



When to Do a Pilot Study

- **After the conceptual model is ready but before full-scale coding or experiments.**
- Essentially, **once you know what components your model needs**, but you want to test feasibility.
- **Example Workflow in AI/ML Paper:**
- Conceptual model: “Multimodal sentiment classification with cross-modal attention and LLM embeddings.”
- Pilot: Train **small model** on 5–10% of dataset, check:
 - Model convergence
 - GPU usage
 - Early metric trends (accuracy/F1)
 - Any preprocessing issues
- Adjust the model based on pilot → proceed to full-scale implementation.

How Pilot Study Helps in Your Paper

- **Validates novelty** → shows method is workable before claiming contribution.
- **Supports methodology section** → justifies hyperparameters, dataset choices, and preprocessing.
- **Reduces risk** → avoids failed experiments at full scale.



Take Aways

- **A pilot study is done after you have planned your research questions**, not during idea generation.
- **Idea/gap finding phase:** You identify problems and possible research directions.
- **Design questions phase:** You formulate specific, testable research questions.
- **Pilot study:** Once questions and conceptual model are ready, you do a **small-scale test** to check feasibility before full experiments.
- **Sequence:** Gap → Idea → Research Questions → Conceptual Model → **Pilot Study** → Full Implementation

Results and Analysis: Datasets, Experiments, and Ablation Studies



1. Purpose of Results Section

- Demonstrate how your method performs relative to:
 - **SOTA baselines** identified from your SLR.
 - **Previous methods** in literature.
- Provide evidence for the validity and novelty of your contribution.
- Highlight **strengths, limitations, and insights.**
- **Most Important:**

“Results are not just numbers they tell the story of whether your method advances the field.”



2. Datasets

- **Selection:**
 - Choose datasets that align with your research problem.
 - Ensure they are **publicly available** (for reproducibility).
 - Mention **size, domain, and features**.
 - **Examples in AI/ML:**
 - **Text:** IMDB, Amazon Reviews, SemEval ABSA datasets
 - **Multimodal:** CMU-MOSEI, MOSEAS, VQA datasets
 - **Images:** CIFAR-10/100, ImageNet
 - **Graphs:** OGB datasets, citation networks
 - **Dataset Split:**
 - Training / Validation / Test (e.g., 70/15/15)
 - Optional: Cross-validation folds
 - Dataset Visualizing: Dataset distribution (bar chart, class balance).

3. Experimental Setup

- **Hardware & Environment:**
 - GPUs/TPUs used, RAM, OS, frameworks (PyTorch, TensorFlow)
 - Seed setting for reproducibility
- **Hyperparameter Tuning:**
 - Learning rate, batch size, optimizer type, number of epochs
 - Use grid search, random search, or Bayesian optimization
- **Baselines:**
 - Include top SOTA methods identified from your SLR.
 - Justify why these baselines were selected.



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Model	Dataset	Accuracy/F1	Notes
BERT	IMDB	91.2%	Standard fine-tuning
MAG-BERT	SemEval ABSA	92.5%	Multimodal baseline
Proposed Model	MOSEAS	94.1%	With cross-modal attention



4. Evaluation Metrics

- Choose metrics based on task:
- **Classification:** Accuracy, Precision, Recall, F1-score, MCC
- **Regression:** MAE, RMSE, R^2
- **Generation/Language Models:** BLEU, ROUGE, METEOR, CIDEr
- **Ranking:** MRR, nDCG
- Report **mean $\pm$ standard deviation** across multiple runs.
- **$\pm$:** It shows the average (mean) and how much the results vary (standard deviation).
- **Full Example:**
 - **Accuracy:** 0.85 ± 0.02
 - **Example**
 - **Means:** You ran the model **5 times** → got accuracies: 0.83, 0.86, 0.84, 0.87, 0.85
 - **Mean = 0.85**
 - **Standard deviation = 0.02**
 - So: **results are around 0.85, usually between 0.83 and 0.87**
 - **Accuracy:** 0.85 ± 0.02
 - **Tip:** Include **metric justification**: Why this metric shows meaningful performance.

5. Ablation Studies

- Purpose: Understand **impact of each component** in your model.
- **Steps:**
- Remove one component (e.g., cross-modal attention) → measure drop in performance.
- Change feature representation → observe effect on metrics.
- Compare model variants to isolate contributions.
- **Visuals:**

– Bar chart showing performance drop for each removed component.

- **Example Table:**

Model Variant	Accuracy	F1-score
Full Model	94.1 ± 0.3%	92.7 ± 0.4%
Without Cross-modal Attention	91.3 ± 0.6%	89.5 ± 0.7%
Without Pretrained Embeddings	90.8 ± 0.8%	88.9 ± 0.9%

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6. Comparative Analysis

- Compare with SOTA models identified in SLR.
- Highlight where your model **improves performance** and discuss why.
- Discuss **limitations** (e.g., computational cost, dataset bias).
- Line chart or bar graph showing SOTA vs your model across multiple datasets or metrics.



7. Error Analysis

- **Error Analysis**
- Identify **common failure cases**
(misclassified examples, noisy data).
- Provide insights into **why the model fails**.
- Suggest improvements in discussion/future work section.
- **Example:**
- Multimodal sentiment misclassifications occur when text and image cues contradict each other